MILITARY SPECIFICATION

EXPLOSIVES; BLASTING AGENT, SLURRY, WITH BOOSTERS, MILITARY CONSTRUCTION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope This specification covers a slurry blasting agent (SBA) for use in breaking rock formations by loading bore holes and blasting for quarry and grade construction excavations in wet and dry, cold (-20 degree Fahrenheit (°F) storage environments) (40°F minimum in bore holes) and warm (+120°F) storage and ambient environments. This document also covers the booster charges required for effective initiation of the column charge of SBA.

1.2 Classification. The SBA and boosters shall be of one type in three sizes.

SBA:

Type I - SBA for use in wet and dry formations from 40°F and above to 120°F.

Size 02 - 2 3/4-inch diameter (16-inch long (approximately)).
Size 03 - 3-inch diameter (16-inch long (approximately)).
Size 05 - 5-inch diameter (29-inch long (approximately)).

Boosters:

Type I - Cast (Commercial) (see table II).

Size 01 - 1/2 pound (lb).
Size 02 - 3/4 lb.
Size 03 - 1 lb.

1.2.1 Specification Part Number (SPN). A SPN system has been developed to aid in the identification of products for acquisition and cataloging. The SPN consists of this specification identifier (M29236) and the SPN code number selected from table I or II (see 6.2). See 6.3 for an example.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified (see 6.2), the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

STANDARDS

FEDERAL

FED-STD-313 - Material Safety Data Sheets Preparation and the Submission of.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-650 - Explosive: Sampling, Inspection and Testing.
MIL-STD-810 - Environmental Test Methods, and Engineering Guidelines.
MIL-STD-1168 - Lot Numbering of Ammunition.
MIL-STD-1751 - Safety and Performance Test for Qualification of Explosives (USAF).

SPECIFICATIONS

MILITARY

MIL-C-14003 - Caps, Blasting, Special.

U. S. AIR FORCE REGULATIONS

AFCS/AFLC Regulation 127-101 - Hazard Classification of Explosives.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

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CODE OF FEDERAL REGULATIONS (CFR)

27 CFR Parts 1-199 - Alcohol, Tobacco and Firearms.
30 CFR Parts 1-199 - Mine Safety and Health Administration.
49 CFR Parts 100-177 - Transportation.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Standard commercial product. The blast materials shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the blast materials being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.2 First article. When specified (see 6.2), the contractor shall furnish blast materials for first article inspection and approval (see 4.2.1).

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.4 Interchangeability. All blasting materials of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies and accessories.

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3.5 Type. One type of SBA shall be furnished. All blasting materials shall be tested to meet the requirements of this section.

TABLE I. Type I SBA Characteristics (Partial).

Size	Dia In. 1/	Length Inch	Lb ea 1/	VOD FPS	Detonator (minimum)	IME Fume class	SPN code No.
02	2-3/4	16	4			3	02
03	3	16	4.7	2/	3/	3	03
05	5	29	25			3	05

1/ Dimensions and weight nominal.
2/ VOD nominal 13,000 feet per second (fps) in 3-inch column, charge, 17,000 fps in 5 inch or larger column charge, confined.
3/ Special booster required (see 3.5.2.5).

3.5.1 Use and description. Type I SBA is for use in loading bore holes for blasting homogenous rock of medium to high seismic velocity range (11,000 fps to over 17,000 fps) with densities of 2.6 through 3.0 grams per cubic centimeter (g/cc). The SBA shall consist of an oxidizer and a sensitizer (combustible fuel) mixed with an aqueous solution and other energy producing materials to produce a thick semifluid mixture. The cartridges shall be cylindrical in shape and wrapped in tear-resistant polypropoylene or plastic film laminate. The ends of the wrapper shall be tied or sealed closed. The wrapper shall be marked to indicate diameter, length, weight, manufacturer's name, explosive name, type, and fume class. When specified, special wrapping shall be provided (see 6.2).

3.5.2 Properties. Specific properties of the SBA are required for storage, transportation, and actual use to ensure reliability, dependability and safety.

3.5.2.1 Storage class. The SBA shall be classifed as a Blasting Agent N.O.S., conforming to 49 CFR part 172.101 and be suitable for storage as a blasting agent in compliance with 27 CFR part 55.

3.5.2.2 Transportation. The SBA shall be classified in hazard class of Blasting Agent N.O.S. for transportation conforming to 49 CFR part 172.101.

3.5.2.3 Fume class. The SBA, when detonated, shall not produce more than 0.67 cubic feet (ft^3) of poisonous or noxious gas per 200 grams of product and be classed as class 3 nonpermissible or better in conformance with 30 CFR part 15 Mine Safety and Health Administration (MSHA) requirements.

3.5.2.4 Density and energy. The SBA density shall be a minimum of 1.20 g/cc specific gravity. The energy available relating to work produced is expressed by several factors; absolute weight strength (AWS), absolute bulk

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strength (ABS), and relative bulk strength (RBS), as compared to the standard ammonium nitrate and fuel oil (ANFO). ANFO is recognized as having an AWS of 890 calories per gram (cal/g), ABS of 730 calories per cubic centimeter (cal/cc), RBS of 100, and a specific gravity of 0.82 g/cc. The SBA shall have a AWS of 800 cal/g minimum (min), ABS of 1000 cal/cc min, RBS of 130 min. The calculated detonating pressure shall be 1.2×10^6 pound-force per square inch (psi) min (see 6.6).

3.5.2.5 Velocity of detonation (VOD). The SBA, in a column charge, shall have a minimum VOD of 13,000 fps when confined in a 3-inch diameter hole and 17,000 fps when confined in a 5-inch diameter hole. The velocity shall be realized with properly primed boosters of 20,000 fps VOD placed at intervals of 10 (ft) (±1 ft) for a 3 inch hole or at intervals of 15 ft (±1 ft) for a 5 inch hole. The minimum column VOD shall be accomplished when boosters are initiated by 18 grain/ft detonating cord (det cord) downline connected to 25 grain/ft det cord threaded through or double half-hitched around each booster, or a military special or commercial equivalent (see table II note 3) electric blasting cap inserted in each booster.

3.5.2.6 Effects of varying storage temperature, and pressures. The SBA performance of 3.5.2.5 shall not have been affected by storage in temperatures of -20°F and +120°F for 12 months or when subjected to rapid decompression to 2.7 psi. The SBA shall not freeze at -20°F and shall not exude, leak, melt or separate after being subjected to these storage conditions.

3.5.2.7 Initiation sensitivity. The SBA shall be insensitive to initiation when subjected to the blasting cap sensitivity and impact sensitivity tests of 49 CFR 173.114a, but shall detonate when initiated as prescribed in 3.5.2.5. The SBA sensitivity shall not be affected by varying ambient pressures from 14.7 psi to 2.7 psi as maybe the case if transported in high flying aircraft.

3.5.2.8 Minimum temperature performance. The SBA shall reach the minimum VOD when the confined column charge temperature has stabilized at 40°F and then detonated with the prescribed primed booster(s) (see 3.5.2.5).

3.5.2.9 Water resistance. The SBA shall perform as specified in 3.5.2.5 after a column charge, with wrapper slit, has been loaded in a water flooded bore hole with 32 (psi) water pressure for 24 hours. The surface of the column charge shall have been in direct contact with the water to determine resistance. A wet hole is meant to mean that ground water seeps into the bore hole past the full length of the column charge, to 32 psi on the charge, becomes static, and retains the 32 psi due to the water table elevation.

3.5.3 Boosters. One type of booster in three sizes are specified to detonate the SBA charges (see 3.5.2.5). Boosters shall not contain nitroglycerin. The booster shall detonate when initiated by a military special blasting cap (instant or delay), or by 25 grains per ft det cord threaded through the booster. Each booster shall provide cap wells and det

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cord tunnels. Cap wells shall be adequate to accept completely long period delay blasting caps. Detonating cord tunnels shall extend through the booster (see table II).

TABLE II. Type I Booster Characteristics (partial).

Size	Dia. In. 1/	Length inch	Weight lb 1/	VOD FPS 1/	Initiator required	Explosive class	Fume class	SPN code No.
01	1-3/4	4-13/16	1/2	26,000	3/	A	2/	11
02	1-15/16	4-13/16	3/4	26,000	3/	A	2/	12
03	2-15/16	4-13/16	1	26,000	3/	A	2/	13

1/ Dimensions, weight and VOD nominal.
2/ Fume class 1 or 2 permitted.
3/ The initiator shall be a military special conforming to MIL-C-14003 or the commercial equivalent. Even though the booster may detonate most times with a No. 8 cap, the military special is specified for dependability.

3.5.3.1 Storage class. The boosters shall qualify for classification as high explosives for storage purposes as defined in 27 CFR part 55.202.

3.5.3.2 Transportation class. The boosters shall be no more hazardous or shock sensitive than class A type 3 explosives as defined in 49 CFR part 173 for commercial boosters.

3.5.3.3 Fume class. Fume class shall be 1 or 2 permitted as rated by MSHA.

3.5.3.4 Velocity of detonation. The booster shall detonate at a velocity of at least 25,000 ft per second when initiated as specified in 3.5.2.5.

3.5.3.5 Resistance to detonate. The booster shall not detonate when initiation is attempted by a number 6 blasting cap. Detonation shall not be affected by temperatures from 120°F to -20°F.

3.5.3.6 Water resistance. Booster performance shall not be affected when immersed in water at 32 psi for a minimum of 24 hours.

3.5.3.7 Environmental effects. The performance of the boosters shall not have been affected by 12 months storage in fluctuating temperatures of -20°F to 120°F, and when subjected to rapid decompression from 14.7 psi to 2.7 psi such as may be the case if the explosives were transported in high flying aircraft and the air craft lost cabin pressure.

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3.5.3.8 Wrapping. The boosters shall be suitably wrapped to prevent loss of granular material from the cast booster. The wrapper shall be marked to indicate diameter, length, weight, manufacturer's name, and booster's name and type and fume class.

3.6 Instructions. Each carton or case of SPA and booster shall contain the "DO's" and "DONT'S" instructions and warnings. Recommendations on proper priming of boosters and placement of the primed booster in the column charge of SBA as described in 3.5.2.5 shall be included.

3.7 Hazardous classification data. As specified (see 6.2 and 6.4), material safety data sheets shall be prepared and submitted in accordance with FED-STD-313, or MIL-STD-1751 (USAF) and AFSC/AFLC Regulation 127-101.

3.7.1 Recipients of data. Material safety data shall be submitted to addressees as specified in the contract (see 6.2 and 6.4).

3.8 Compliance. Prior to approval of the first shipment, the contractor shall submit to the contracting officer or authorized representative, satisfactory evidence that the explosives the contractor proposes to furnish under this specification meets the requirements of section 3 of this specification as applicable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use owned or other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Several aspects of the inspection maybe extremely hazardous to life and property, therefore the necessary precautions shall be observed.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on the SBA and boosters to be delivered on a specific contract or purchase order when a first article is required (see 3.2 6.2, and 6.5). This inspection shall include the examination of 4.4 and the tests of 4.5 and 4.6. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets

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the requirements of this specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract. When a first article is not performed, a document from an independent laboratory acceptable to the US Government, shall be provided certifying that the product provided conforms to this specification.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, the tests of 4.5 through 4.6.7, and the packaging inspection of 4.7 as applicable. This inspection shall be performed on the samples selected in accordance with 4.3 through 4.3.4. Data collected during examinations and tests of 4.4 thru 4.5.6 shall be recorded and retained for later use as will be required for comparison to results of the environmental effects tests of 4.5.7 and 4.6.7.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105.

4.3.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same specification and same specification revision. Specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.2 Inspection lot. An inspection lot shall consist of all explosives (SBA and boosters) of the same classification represented in a production lot. The unit of product shall be one unit of issue as defined in the contract or purchase order.

4.3.3 Sampling for examination. Examination shall be based on inspection level II and an Acceptable Quality Level (AQL) of 0.65 percent defective for major defects and 0.40 percent defective for minor defects.

4.3.4 Sampling for tests. Tests shall be based on inspection level S-4 and an AQL of 0.40 percent defective. Samples for tests shall be selected from lots which were examined and accepted.

4.4 Examination. Each sample selected shall be examined for defects listed in table III.

TABLE III. Classification of defects.

Classification	Defects	Requirement paragraph
Critical:	Any defect affecting the safety of personnel handling the explosive.	None
Major:		
101	Items of different manufacturers, of different type and styles not interchangeable for the intended purpose.	3.4

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TABLE III. Classification of defects (con't).

Classification	Defects	Requirement paragraph
102	Dimensions not as specified.	Table I
103	Weight not as specified.	Table I
104	Explosives which have been subjected to equivalent 12 months storage warm-cold conditions (cycled) (see 4.6.4) shall not show evidence of explosive ingredients or other components leaking, exuding, melting, flowing, or separating from the explosive.	3.5.2.6
105	Identification not as specified.	3.5.1
106	Cap wells not adequate to accept completely long period delay blasting caps.	3.5.3
107	Detonating cord tunnels not extending through the boosters.	3.5.3
108	Dimensions not as specified.	Table II
109	Weight not as specified.	Table II
110	Wrapper not as specified.	3.5.3.8
111	Identification not as specified.	3.5.3.8
Minor:		
201	Wrapping ends not tied or sealed.	3.5.1
202	Other wrapping not as specified.	3.5.1
203	Instructions not as specified.	3.6
204	Evidence of compliance.	3.8

4.5 Tests, SBA. Tests of 4.5.1 through 4.5.10 shall be conducted to determine compliance with Section 3.

4.5.1 Classification for storage. To determine compliance with 3.5.2.1, the SBA shall be tested as specified in 49 CFR part 173.114a to ensure suitability for storage as a blasting agent as defined in 27 CFR part 55.

4.5.2 Classification for transportation. To determine compliance with 3.5.2.2, the SBA shall be tested as specified in 49 CFR part 173.114a.

4.5.3 Fume class. To determine compliance with 3.5.2.3, the SBA shall not produce more than 0.67 cubic ft of poisonous or noxious gases per 200 grams sample of SBA when tested as specified in 30 CFR part 15 (MSHA).

4.5.4 Density and energy. To determine compliance with 3.5.2.4, the SBA shall be tested for specific gravity and AWS, and calculated for ABS, RBS, and detonating pressure (DP) (see 4.5.4.1 through 4.5.4.5 and 6.6).

4.5.4.1 Specific gravity. Perform test method 203.2 of MIL-STD-650 (see 3.5.2.4).

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4.5.4.2 Absolute weight strength. Conduct laboratory tests and evaluations of all components of the SBA to determine heat-of-formation that will contribute to explosive energy. Express the AWS as cal/g. Conduct the same test with the same weight of standard ANFO. The test should indicate an AWS of 890 cal/g (±5 percent) for the ANFO. The test with the ANFO is to determine the validity of the lab evaluation of the SBA.

4.5.4.3 Absolute bulk strength. Calculate the ABS by the formula: ABS = (AWS) (Density). Express the ABS in cal/cc.

4.5.4.4 Relative bulk strength. Calculate the RBS by the formula: RBS = (ABS of product in question) divided by (ABS ANFO) and multiply the quotient by 100.

4.5.4.5 Detonating pressure. Calculate the DP by the formula: DP in psi = (0.0363) (Density in g/cc) (Velocity in meters)^2. DP in kilobars (kbar) = (psi) divided by (14,504 psi/kbar).

4.5.6 Velocity of detonation. Perform test method T109.1 of MIL-STD-650 (as modified herein) to determine VOD (see 3.5.2.5). Maintain record of dimensions between det cord on the charge and distance from center of det cord to impact mark on plate for comparison after environmental tests, minimum temperature, and water resistance tests (see 4.5.7, 4.5.9 and 4.5.10). Modify the procedure by confining the charge in a steel pipe with ends capped. Drill holes in end caps and sides of pipe to insert detonator and det cord.

4.5.7 Environmental effects. The tests of 4.5.7.1 through 4.5.7.3 are required to determine effects of the environment on the explosive materials relative to sensitivity for detonation, VOD and deterioration as addressed in Section 3. The information produced and collected during 4.4 through 4.5.6 will be required to compare to the results of these tests to determine compliance with Section 3.

4.5.7.1 Effects of high temperature. Subject the sample(s) (2 lb min) to test method 501.2 of MIL-STD-810.

Test conditions:

Procedure: Number 1
Climate category: H-1 (Hot) with covered ventilated storage.
Duration: 7 cycles.
Data: Collect information as evidenced during the test (leaking, exuding, crumbling, color and odor change, etc.)

Upon completion of test 501.2, conduct the hazard class and velocity tests of 4.5.1 and 4.5.6 and compare results with pre-recorded data (see 4.2.2) to evaluate changes to determine compliance with Section 3 for the class required such as:

a. Pass No. 8 blasting cap test.
b. Pass differential thermal analysis test.

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c. Pass thermal stability test.
d. Pass electrostatic sensitivity test.
e. Pass impact sensitivity test.
f. Pass fire test.
g. VOD not reduced.

4.5.7.2 Effects of low temperature. Subject the sample(s) (2 lb min) to test method 502.2 of MIL-STD-810.

Test conditions:

Test: Procedures I and II.
Climate category: Basic Cold (C1)
Duration: 72 hours min in storage.

Upon completion of test 502.2, conduct the hazard class tests of 4.5.1. Compare results with pre-recorded data (see 4.2.2) to evaluate changes to determine compliance with Section 3 for the class required.

4.5.7.3 Effects of atmospheric pressure changes. Subject the sample(s) (2 lb min) to test method 500.2 of MIL-STD-810.

Test conditions:

High pressure:	14.7 psi
Low pressure:	2.7 psi
Decompress time:	15 seconds maximum (max)
Recompress time:	5 minutes max
Test procedure:	Procedure II (operation)
	Procedure III (Rapid decompression)
Test procedure sequence:	Procedure III before II.
Duration:	Time required to complete functions.

Upon completion of both procedures, conduct hazard class tests of 4.5.1 to determine compliance with Section 3 such as:

a. Pass No. 8 blasting cap test.
b. Pass differential thermal anaylsis test.
c. Pass thermal stability test.
d. Pass electrostatic sensitivity test.
e. Pass impact sensitivity test.
f. Pass fire test.

4.5.8 Initiation sensitivity. The SBA shall not detonate when subjected to the blasting cap sensitivity, impact sensitivity and fire test of 49 CFR 173.114a (see 3.5.2.7).

4.5.9 Performance at minimum temperature. Properly primed, a 3 inch diameter 10 ft column charge of -20°F SBA confined shall detonate when the SBA temperature stablizes at +40°F. The column shall be loaded with SBA that had

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been in -20°F storage for the time required to cool the SBA throughout. The test method T109.1 of MIL-STD-650 (see 4.5.6) shall be set up with the column charge to check VOD. The column shall be detonated by primed boosters, as prescribed in 3.5.2.5 (see 3.5.2.8). The charge shall detonate completely at not less than the velocity of 3.5.2.5.

4.5.10 Water resistance. Place a min of 2 lb confined charge of SBA in a container of water of appropriate size and produce a pressure head equivalent to 32 psi (73.8 ft H_2O). Maintain this condition for 24 hours. The charge shall be prepared for detonation prior to placing in the test situation. The primed booster used for detonation shall be as prescribed in 3.5.2.5. Detonation of the charge is to determine compliance with 3.5.2.9. Detonation shall be attempted without removing the charge from its water environment.

4.6 Tests, booster. The tests of 4.6.1 through 4.6.7 shall be conducted to determine compliance with Section 3 of this specification. Record information produced during the tests of 4.6.1 thru 4.6.5 for comparison with results of environmental effects tests of 4.6.7.

4.6.1 Storage class. Conduct the tests of 49 CFR part 173 for class A type 3 to determine if the boosters conform to requirements to be classed as high explosives for storage as defined in 27 CFR part 55.202 (see 3.5.3.1).

4.6.2 Transportation class. Determine compliance with 3.5.3.2 by conducting the impact test of 49 CFR part 173.53(c), and the safety fuse flame spit test. To conduct the flame spit test, pack a dry sample of pulverized booster material into a 6 inch section of glass tube (of inside diameter that fits snuggly over safety fuse) to within 1/2 inch of the open end and insert a squared end of safety fuse (24 inch long min). Tape the safety fuse to the glass tube to hold the safety fuse against the compacted booster material, ignite the safety fuse and observe closely to ensure that the flame spit from the fuse actually occurs. If the test sample flames, smolders or glows from the flame spit, the material cannot be considered class A type 3.

4.6.3 Fume class. To determine compliance with 3.5.3.3, the booster shall not produce more than 0.33 cubic ft of poisonous or noxious gases per 200 grams sample of booster material when tested as specified in 30 CFR part 15 (MSHA).

4.6.4 Velocity of detonation. To determine compliance with 3.5.3.4, perform test method T109.1 of MIL-STD-650. Conduct the test on a ft long (min) column of boosters held tightly butted together in a paper tube. Maintain record of dimensions used during the test for future comparisons with results of environmental tests.

4.6.5 Resistance to detonate. To determine compliance with 3.5.2.5 and 3.5.3.5, attempt to detonate the booster with a No. 6 blasting cap properly inserted in the cap well, and attempt to detonate the booster with a military special blasting cap properly inserted in the cap well.

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4.6.6 Water resistance. Determine compliance with 3.5.3.6 by conducting the test of 4.5.10 either in conjunction with testing the SBA or by itself. A min of 1/2 lb of booster sample shall be used for this test.

4.6.7 Environmental effects. Conduct the tests of 4.6.7.1 and 4.6.7.3 to determine effects of exposure to fluctuating temperatures between -20°F and 120°F and the effects of rapid change in pressures. The military special cap is used in case the booster becomes less sensitive to shock (see 4.6.7.1 thru 4.6.7.3).

4.6.7.1 High temperature. To determine compliance with 3.5.3.7, conduct the test of 4.5.7.1 with a min sample of 1/2 lb. Upon completing all cycles, and data is recorded, conduct the tests of 4.6.2 and 4.6.5. Should the booster indicate increased sensitivity to detonate or burn, or fail to detonate with military special cap, the lot does not meet requirements.

4.6.7.2 Low temperature. To determine compliance with 3.5.3.7, conduct the test of 4.5.7.2 with a min sample of 1/2 lb. Upon completing of the test, conduct the tests of 4.6.2 and 4.6.5. Should the booster indicate increased sensitivity to detonate or burn, or fail to detonate as required with a military special cap, the lot does not meet requirements.

4.6.7.3 Pressure changes. To determine compliance with 3.5.3.7, conduct the test of 4.5.7.3 with a min sample of 1/2 lb. Upon completion of both procedures, conduct the tests of 4.6.2 and 4.6.5. Should the booster indicate increased sensitivity to detonate or burn, or fail to detonate as required with a military special cap, the lot does not meet requirements.

4.7 Packaging inspection. The preservation, packing, and marking of the item shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation and packing. Unless otherwise specified (see 6.2), the preservation and packing of material shall be in conformance to the applicable carrier rules for transporting explosives.

5.2 Marking. Marking shall be in accordance with MIL-STD-129 to include the shelf-life marking.

6. NOTES

6.1 Intended use. The SBA and boosters are intended for use by the Department of Defense for breaking up hard consolidated earth or homogeneous rock in quarries and grade construction.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When the current issue of reference documents are not to be used and a different issue is to be used (see 2.1.1).

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c. SPN of explosives required (see 1.2.1, 6.3, tables I and II).
d. When special wrapping are required (see 3.5.1).
e. When a first article is required for inspection and approval (see 3.2, 4.2.1 and 6.5).
f. Standard requiring safety data sheets (see 3.7).
g. Recipients of safety data sheets (see 3.7.1).
h. Packaging requirements, if different (see 5.1).

6.3 The SPN consists of the identifier of this specification (M29236) plus the SPN code number from tables I or II (extreme right column).

Example - M29236-01 indicates that size 02 , type I SBA cartridge is required.

M29236 - 01
Specification identifier
SPN code from right column of table I

6.4 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of paragraph 52.227-7031 of the Federal Acquisition Regulations (FAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of FAR 52.227-7031 are not invoked, the data shall be delivered in accordance with the contract requirements.

6.5 First article. When a first article inspection is required, the item will be tested and should be a first production item, or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.6 Calculation of SBA energy. Table III indicates the required energy availability of SBA compared to the nationally recognized energy of ANFO.

TABLE III. Comparision of ANFO and SBA Energy.

	ANFO	SBA
Density	0.82 g/cc	1.2 to 1/25 g/cc
AWS	890 cal/g	800 cal/g
ABS	730 cal/cc	1000 cal/cc
RBS	100	130 (min)

6.6.1 Formula for absolute bulk strength. (ABS) = (AWS) (Multiplied by the density of sample in g/cc).

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6.6.2 Formula for relative bulk strength. (RBS) = (ABS product "X") divided by (ABS ANFO) (Multiplied by 100).

6.6.3 Formula for detonating pressure (DP). (DP in psi) = (0.0363) (Multiplied by the density in g/cc) (Multiplied by the square of the velocity in meters).

(DP in kbars) = (DP in psi) divided by (14,504 psi/kbar). The DP (brisance) is the fraction-of-a-second shock wave that exists to produce the shattering effect that fatigues the rock for size control prior to the heaving action of the longer duration gas expansion (bore hole pressure).

Custodians:

Army - CE
Navy - YD
Air Force - 99

Review Activity:

Navy - OS

User Activity:

Navy - MC

Preparing Activity:

Navy - YD

(Project 1375-0280)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-E-29236	EXPLOSIVES: BLASTING AGENT, SLURRY, WITH BOOSTERS MILITARY

3a. NAME OF SUBMITTING ORGANIZATION CONSTRUCTION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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